PigMENT: A Web-Based Swine Growth Monitoring, Marketplace, and Disease Awareness System for Small-Scale Pig Farmers in Marinduque

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ABSTRACT

Small-scale pig farmers in Marinduque need practical ways to maintain livestock records, estimate live weight when weighing scales are unavailable, share pigs for sale, and access disease-related information. This project developed PigMENT, a web-based system that brings these functions together. The team used an iterative and incremental development approach and built the system with HTML, CSS, JavaScript, PHP, and MySQL. Its three core features are pig growth monitoring and weight estimation, a community livestock marketplace, and pig health information with geographic disease awareness. During functional testing, the system generated live-weight estimates from heart girth and body length, displayed growth records, and supported pen assignments and transfers. The marketplace allowed users to create, search, filter, and view livestock listings. The disease awareness feature accepted suspected disease reports with location details, supported administrative review, and displayed approved disease information on an interactive map. These results showed that the three core workflows operated as intended under the conditions tested. However, the evaluation did not establish the accuracy of weight estimates against scale readings or the system's effectiveness in actual farming operations. PigMENT supports disease awareness and reporting but does not diagnose disease or independently confirm cases.

Keywords: agriculture and food innovation, livestock marketplace, web-based system, swine growth monitoring, disease awareness

I. INTRODUCTION

1.1 Background and Problem Context

In Marinduque, the swine industry faces production and market-related challenges that affect local pig farmers and the province's pork supply. The provincial government of Marinduque reported rising pork prices and declining pork supply in the province. The report identified high animal feed costs, increasing production expenses, and the recent African Swine Fever (ASF) outbreak as factors affecting local hog production. In response, the provincial government coordinated with stakeholders in the pork supply chain to develop measures addressing these concerns. These developments highlight the existing challenges within Marinduque's swine industry and the importance of supporting local pig farmers [18]. Furthermore, a study on Philippine native pigs demonstrated the potential of using external body measurements, particularly heart girth and body length, to estimate liveweight [1]. This supports the possibility of providing small-scale farmers with an accessible method of monitoring pig growth, especially when weighing scales are not readily available.

Disease prevention and awareness are also important concerns in Marinduque's swine industry. The Marinduque Provincial Veterinary Office strengthened ASF monitoring at the province's ports following suspected cases in nearby provinces. The provincial government also implemented restrictions on the entry of live pigs and pork products to help protect the province's ASF-free status. These preventive measures emphasize the importance of accessible and reliable disease information for pig farmers, barangay officials, and agricultural authorities. P

Although existing reports and studies establish challenges affecting Marinduque's swine industry and smallholder pig production in the Philippines, the specific digital recordkeeping, livestock marketing, and disease information needs of small-scale pig farmers in Marinduque have not yet been sufficiently established. This research gap highlights the need for local assessment to determine how these concerns affect farmers throughout the province.

To address these potential needs, the proposed PigMENT system is a web-based platform designed to support pig growth monitoring, digital pen and livestock record management, community livestock listings, and access to pig health information and geographically presented disease reports. The system will be evaluated to determine its usability and acceptability as a digital management and information tool for small-scale pig farmers in Marinduque.

1.2 Problem Statement and Research Gap

Small-scale pig farmers need to monitor pig growth, maintain farm records, estimate selling prices, and access information that can support their farming activities. However, estimating the live weight of pigs may be challenging when weighing scales are not readily available. Most farmers rely on 'eyeball' estimation to market pigs. Previous studies have shown that external body measurements, such as heart girth and body length, can be used to estimate pig live weight. Other studies have also explored digital tools for organizing pig records and monitoring livestock information, demonstrating how technology can support swine farm management [1], [2], [5], [7], [11], [20].

Access to market information is another important consideration in livestock production. Existing studies provide a basis for considering an online marketplace where users can post livestock listings and view relevant information, such as breed, estimated weight, price, and location [4], [13], [14], [18]. Examined pig farmers' biosecurity practices, disease awareness, and reporting of suspected African Swine Fever (ASF) cases, highlighting the importance of accessible disease information and appropriate reporting practices [6], [10], [15], [16], [17]. In Marinduque, government reports have discussed efforts to strengthen ASF monitoring and address concerns related to pork supply and hog production, showing the local relevance of livestock health and market coordination [3], [18].

Although existing studies and systems have addressed pig weight estimation, farm record management, agricultural marketplaces, and disease-related information, these functions are often examined or implemented separately. This presents an opportunity to develop a single platform that brings selected functions together for small-scale pig farmers. The specific research gap addressed by this project is the development of an integrated web-based system that combines pig growth monitoring, a livestock marketplace, and disease awareness features within the context of Marinduque. However, the reviewed literature and available government reports do not show how common these difficulties are among small-scale pig farmers throughout the province.

To address this gap, the project proposes **PigMENT: A Web-Based Swine Growth Monitoring, Marketplace, and Disease Awareness System for Small-Scale Pig Farmers in Marinduque**. The system will allow users to record pig information, estimate live weight using selected external measurements, track growth history, calculate estimated selling value, browse livestock listings, and access disease-related information and geographic awareness features. Reports of suspected disease cases will be subject to administrative review, and the system will not independently diagnose diseases or confirm outbreaks.

1.3 Objectives

To develop PigMENT, a web-based swine growth monitoring, marketplace, and disease awareness system designed to support small-scale pig farmers in Marinduque in managing pig growth information, accessing livestock listings, and obtaining disease-related information.

Specifically, this project aims to:

1. Develop a pig growth monitoring feature that allows users to record pig information, estimate live weight using selected external body measurements, track growth history, and calculate estimated selling value.
2. Develop a livestock marketplace feature that allows users to post, browse, and view livestock listings containing relevant details such as breed, age, estimated weight, asking price, and general location.
3. Provide a disease awareness feature that presents information about common swine diseases, ASF prevention, and biosecurity practices, along with geographic awareness information and a reporting feature for suspected disease concerns.
4. Implement a web-based database and user interface that organize pig records, marketplace listings, and disease-related information in one platform.
5. Test the core system functions to determine whether the implemented features work according to their intended requirements and identify limitations that may require improvement.

1.4 Proposed Solution, Scope and Limitations

PigMENT is a proposed web-based system for monitoring growth performance, marketplace activities, and disease awareness among small-scale and backyard pig farmers in Marinduque. The system will provide a centralized platform where farmers can maintain livestock information, monitor the estimated growth of their pigs, calculate potential selling values, find buyers, and access relevant livestock disease information. Existing web based livestock management systems have explored ways to support the monitoring of reproductive performance, diseases, and biosecurity in swine farms [10]. Web-based record-keeping applications have also demonstrated their potential to help native pig farmers organize and manage livestock information [20].

The system will focus on three main features. First, the pig growth monitoring and weight estimation feature will allow farmers to register their pigs and record information such as breed, age, sex, heart girth, and body length. These measurements will be used to estimate live weight and maintain a history of recorded growth data. Previous studies have examined the use of external body measurements, particularly heart girth and body length, to estimate pig live weight [1], [2], [8]. Farmers will also be able to enter a price per kilogram to estimate the potential selling value of their pigs.

Second, the community livestock marketplace will allow registered farmers to create livestock listings containing relevant information about their pigs, including estimated weight, age, asking price, and general location. Potential buyers will be able to browse available listings and contact sellers directly. Previous studies on agricultural marketplaces and livestock supply chain platforms have explored how digital platforms can connect agricultural producers with potential buyers and improve access to market information [4], [13], [14].

Third, the pig health information and geographic disease awareness feature will provide basic information about common pig health concerns, African Swine Fever (ASF), preventive measures, and biosecurity practices. Previous studies have examined biosecurity practices, disease recognition, and early reporting among smallholder pig farmers [6], [15], [16].

The study is limited to developing a web-based system intended for small-scale and backyard pig farmers in Marinduque. The system will rely on information manually entered by users and will not require specialized sensors or automated weighing equipment. Unlike systems that use wireless communication for pig growth monitoring [5].

The calculated live weight will serve only as an estimate as its accuracy depends on the selected estimation formula, the characteristics of the pig, and the accuracy of the measurements provided by farmers [1], [2], [8]. Reports submitted by farmers will undergo administrative review, and only authorized veterinary personnel will be able to identify cases as confirmed. These limitations will keep the proposed system focused on its three primary functions: livestock growth monitoring, community marketplace access, and disease awareness.

II. METHODOLOGY

2.1 Development Approach

The development of PigMENT followed an **iterative and incremental development approach**, in which the system was developed and improved through successive implementations of its core features. The team initially created a frontend prototype using HTML, CSS, and JavaScript to establish the system's interface, navigation, and basic functionality. The initial implementation focused on pig registration and management, digital pig pens, and the community livestock marketplace.

As development progressed, the team reviewed and tested the implemented features to identify errors, incomplete functions, and areas requiring improvement. Modifications were made to improve usability, functionality, and consistency across the system. Additional components, including pig growth monitoring, disease reporting, geographic disease awareness, and user account management, were progressively developed. PHP and MySQL were subsequently introduced to support backend processing, user authentication, and persistent data storage. This approach allowed the team to develop the system in manageable components, evaluate existing functionality, and make necessary improvements while integrating additional features. Functional verification was conducted on implemented components to assess whether they performed their intended operations and satisfied the identified system requirements.

2.2 Requirements and Core Features

Table 1. Requirements and Core Features

Problem / need	Core feature	Acceptance criterion
Difficulty organizing and monitoring pig farm records	Pig Growth Monitoring and Weight Estimation	The system successfully allows farmers to register, view, and update pig records; calculate estimated live weight using entered heart girth and body length measurements; display growth history; create and manage digital pens; assign or transfer pigs between pens; and litter information.
Difficulty presenting livestock information and connecting with potential buyers	Community Livestock Marketplace	The system successfully allows farmers to create and manage livestock listings containing relevant pig information, estimated weight, asking price, and location.
Limited access to organized pig health information and disease-reporting channels	Pig Health Information and Geographic Disease Awareness	The system successfully provides access to pig health and biosecurity information, allows farmers to submit suspected disease reports with location details, and enables authorized administrators to review and update report statuses.

2.3 Interface Architecture and Data Handling

PigMENT uses a web-based interface organized into three core features: Pig Growth Monitoring and Weight Estimation, Community Livestock Marketplace, and Pig Health Information and Geographic Disease Awareness. Additional interfaces include the dashboard, user profile, reports, and administrative management.

The system uses HTML, CSS, and JavaScript for interface design and user interactions, including form submissions, calculations, filtering, and navigation. PHP handles server-side processing and communicates with a MySQL database for persistent data storage, while JSON is used to exchange data between the frontend and backend. Leaflet.js and OpenStreetMap support geographic visualization, and Chart.js displays growth records. User access and data modifications are controlled through authentication and role-based permissions.

2.4 Technology Selection and Comparative Justification

Table 2. Technologies and Justification

Technology	Alternative considered	Why chosen for THIS project	Evidence
HTML5	[]	HTML5 provides the foundation for organizing the different sections and input forms of the web application. The web based swine recordkeeping systems in [20] and demonstrate the relevance of organizing livestock information through accessible web interfaces [34]. Technical reference: [26].	[20], [34], [26]
CSS3	[]	The agricultural information systems in [20] and [35] demonstrate the use of web interfaces for presenting farmrelated information. Technical reference: [27].	[20], [35], [27]
JavaScript	[]	The interactive web-based livestock systems in [34] and [36] support the value of interactive information presentation. Technical reference: [28].	[28], [34]. [36]

PHP	[]	PHP is suitable because it has already been used in web-based livestock and agricultural management systems. Parisutthikul et al. [34] developed a swine management information system using PHP and MySQL, while Manabat et al. [35] used PHP in an agricultural monitoring system. Technical reference: [29].	[29], [34], [35]
MySQL	[]	A relational database is appropriate for organizing information that is connected across different modules. Parisutthikul et al. [34] used MySQL for swine breeding herd records, while Abril Santos et al. [25] demonstrated web-based recordkeeping for Philippine native pigs. Technical reference: [30].	[20], [30]. [34]
Leaflet.js and OpenStreetMap	[]	Geographic mapping is relevant to disease awareness because it allows users and authorized personnel to understand where reported incidents are located. Flores [21] and Bulawit et al. [22] explored web-based GIS applications for ASF monitoring and surveillance. Manabat et al. [35] also demonstrated the use of 2D mapping in an agricultural monitoring system. Leaflet provides the mapping functionality, while OpenStreetMap supplies map data. Technical references: [31] and [32].	[25], [17], [31], [32], [35]
Chart.js	[]	Visual representations can make changes in pig measurements and farm performance easier to understand. Parisutthikul et al. [34] incorporated reports and graphs into a swine management system. Morota et al. [37] discussed the importance of interactive graphics in animal science, while Petukhova et al. [36] demonstrated interactive dashboards for livestock disease monitoring. Technical reference: [33].	[33], [34], [36], [37]

E. Implementation and Evaluation Procedure

PigMENT system was created as a web-based system through use of HTML, CSS, JavaScript, PHP and MySQL, with data visualization and disease mapping done by Chart.js and Leaflet.js. The implementation of the system was done based on three major functions namely; Pig Growth Monitoring and Weight Estimation, Community Livestock Marketplace and Pig Health Information and Geographic Disease Awareness. In addition to this, functional testing involved the tests for the record management, weight estimation, community marketplace, disease reporting, administration and maps. The interface testing involved tests of the response of the system to various viewports and keyboard accessibility. The outcome of the test was recorded through use of screenshots and where problems existed, it was fixed and tested again.

III. RESULTS AND DISCUSSION

3.1 Core Feature 1: Pig Growth Monitoring and Weight Estimation

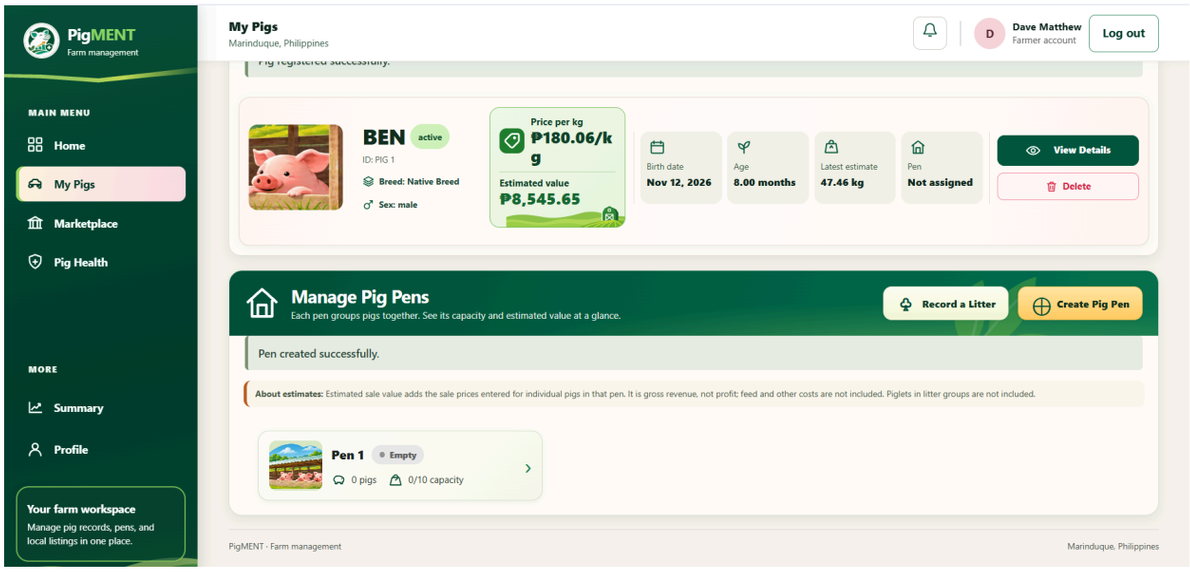


Figure 1. Pig Growth Monitoring and Weight Estimation

The Pig Growth Monitoring and Weight Estimation feature helps farmers organize pig records and monitor livestock growth. Users can register pigs, enter information such as breed, age, sex, heart girth, and body length, and obtain an estimated live weight and selling value. As shown in Figure 1, the interface also supports digital pen management, pig assignment and transfer, growth history, and litter records.

Functional testing confirmed that pig registration, weight estimation, growth tracking, pen management, and litter management worked as intended. However, estimated weights depend on the accuracy of the measurements provided by users.

3.2 Core Feature 2: Community Livestock Marketplace

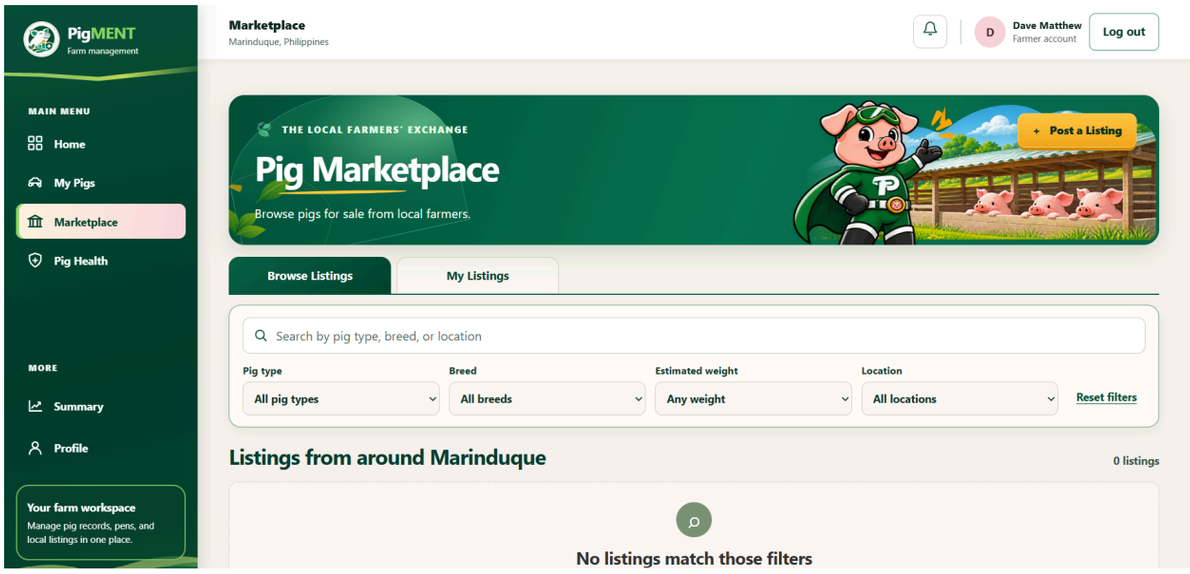


Figure 2. Community Livestock Marketplace

The Community Livestock Marketplace provides a platform for farmers to advertise their livestock and connect with potential buyers. Users can post listings containing pig details such as breed, age, estimated weight, asking price, and location. As shown in Figure 2, the interface includes browsing, searching, filtering, and listing management functions, with seller information available for each listing.

Functional testing confirmed that listing creation, management, searching, filtering, and viewing listing details worked successfully. However, the platform only facilitates connections between buyers and sellers and does not guarantee completed transactions.

3.3 Core Feature 3: Pig Health Information and Geographic Disease Awareness

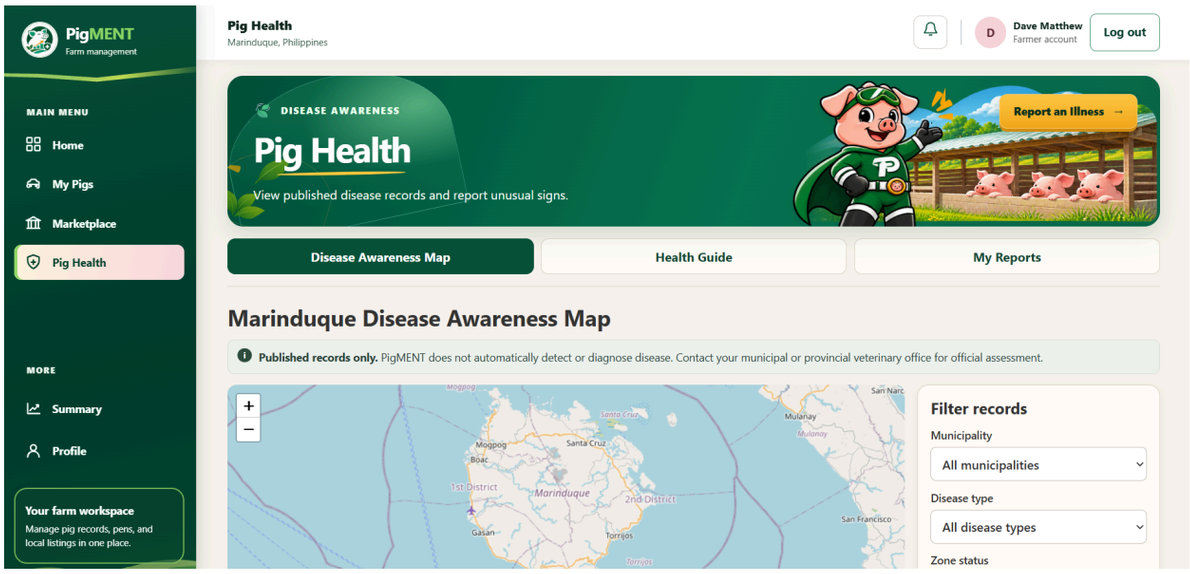


Figure 3. Pig Health Information and Disease Awareness

The Pig Health Information and Geographic Disease Awareness feature provides farmers with information about common pig diseases, African Swine Fever (ASF), and biosecurity practices. As shown in Figure 3, users can access health guides, submit suspected disease reports with location details, and view verified disease-related information through an interactive map. The system also allows administrators to review reports and update their statuses.

Functional testing confirmed that health information, disease reporting, administrative review, and geographic mapping worked as intended. However, the system does not independently diagnose or confirm diseases; submitted reports require verification by authorized veterinary personnel.

3.4 Functional Verification and Limitations

Table 3. Core Functional Features

Core feature	Expected behavior	Observed evidence	Status
Pig Growth Monitoring and Weight Estimation	Farmers can register pigs, create individual pig profiles, record body measurements, view estimated weight and growth history, create digital pens, and assign or transfer pigs between pens. They can also organize sows and their litters.	Pig registration and profile management successfully recorded livestock information. The system calculated estimated weights from entered body measurements, pig assignment and transfer, displayed growth records, and litter management.	Passed
Community Livestock Marketplace	Farmers can create livestock listings containing breed, age, estimated weight, price, and general location. Other users can browse, search, filter, and view available livestock listings and seller information.	Livestock listings were successfully created and displayed with their corresponding details. Search and filtering functions returned matching listings, while the listing details view displayed pig information, asking prices, locations, and seller information.	Passed
Pig Health Information and Geographic Disease Awareness	Farmers can access pig health and biosecurity information, submit suspected disease reports with location details, and view verified disease incidents on an	Pig health information and biosecurity guidelines were accessible through the system. Suspected disease reports were	Passed

	interactive map. Authorized personnel can review reports, update their status, and publish verified incidents.	submitted with location information, while the administrative interface supported report review and status updates. Verified and approved disease information was displayed through the interactive map.

IV. CONCLUSION

PigMENT was developed as a web-based system that brings together livestock recordkeeping, local marketplace listings, and disease awareness for small-scale pig farmers in Marinduque. The system allows farmers to register pigs, manage pens and litter records, and track estimated live weight over time using heart girth and body length measurements. A separate calculator estimates live weight and potential selling value without requiring users to register a pig. The marketplace lets farmers post and manage livestock listings and lets prospective buyers search, filter, and view listing details before contacting sellers directly. The pig health module provides educational information and allows farmers to submit suspected disease reports with location details. Administrators can review these reports, and the public map displays only published incidents and zones that meet the system's confirmation requirements. The functional checks documented in Section 3.4 showed that the three core workflows performed as expected in the scenarios tested.

These capabilities provide farmers with one place to maintain livestock records, share sale information, and access disease updates. However, the functional checks did not establish how closely the weight estimates match actual scale readings or how well PigMENT fits farmers' daily routines. The marketplace does not process payments, and the disease module does not diagnose illness or independently confirm cases. Future work should compare estimated and measured weights for pigs of different sizes, test the system with local farmers using their own devices and internet connections, and consult local veterinary personnel about how the reporting and publication workflow could fit existing practices.

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CRediT AUTHOR CONTRIBUTIONS STATEMENT

Project: PigMENT **Group Name:** Buttercocobonia **Course / Section:** BSI/T 3

Instructions: Mark the relevant roles for all 5–6 members. Multiple people may share a role and any contributor may have multiple roles. Each member should verify their entries. CRediT documents contributions; it does not determine who qualifies for authorship.

Contributor	C	M	So	Va	Fo	I	R	D	O	E	Vi	Su	P	Fu
Author 1: Dave Matthew O. Laman	☑	☑	☑	☑	☑	☐	☑	☐	☑	☑	☑	☑	☑	☐
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Author 3: Lord Byron H. Saguid	☑	☐	☑	☐	☐	☐	☑	☐	☐	☐	☑	☐	☐	☐
Author 4: Ashleigh Mahaguay	☐	☑	☐	☐	☐	☑	☑	☐	☑	☑	☐	☐	☐	☑

ROLE LEGEND

Code	Role	Code	Role
C	Conceptualization	M	Methodology
So	Software	Va	Validation
Fo	Formal analysis	I	Investigation
R	Resources	D	Data curation
O	Writing – original draft	E	Writing – review & editing
Vi	Visualization	Su	Supervision
P	Project administration	Fu	Funding acquisition

Member confirmation:

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/ November 12, 2026.

CRediT role definitions: https://credit.niso.org/contributor-roles/ . Roles marked must reflect substantive actual contributions; no role is mandatory by default.

AI USE AND SUPPORTING EVIDENCE

The group used GitHub Copilot for software development and interface refinement and ChatGPT for manuscript drafting and revision. The supporting records describe the tasks, affected sections or files, exact prompts, outputs used, validation methods, and responsible members where documented. See the AI_DISCLOSURE.md file in the PigMENT GitHub repository for the disclosure log and the accompanying CICS form.